

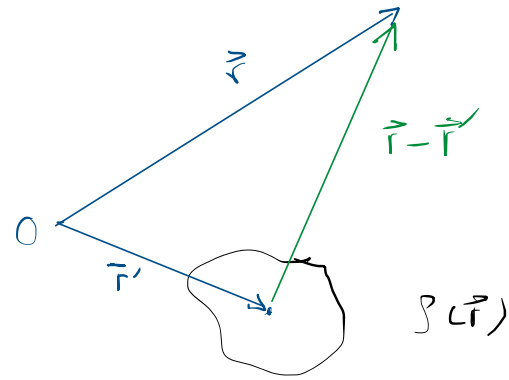
$$-\nabla^2 \psi(\vec{r}) = \frac{1}{\epsilon_0} \rho(\vec{r}) \quad \text{cont. dist.}$$

Point source

$$G(\vec{r}, \vec{r}') = \frac{1}{4\pi} \frac{1}{|\vec{r} - \vec{r}'|}$$

General source

$$\begin{aligned} \psi(\vec{r}) &= \frac{1}{4\pi\epsilon_0} \int d^3r' \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|} \\ &= \int G(\vec{r}, \vec{r}') f(\vec{r}') d^3r' \end{aligned}$$



$$G(\vec{r}_1, \vec{r}_2)$$

$$\Leftrightarrow$$

$$\mathcal{L} G(\vec{r}_1, \vec{r}_2) = \delta(\vec{r}_1 - \vec{r}_2)$$

Green's Function

$$\psi(\vec{r}_1) = \int G(\vec{r}_1, \vec{r}_2) f(\vec{r}_2) d^3r_2 \Leftrightarrow$$

$$\mathcal{L} \psi(\vec{r}_1) = f(\vec{r}_1)$$

$$\begin{aligned} \mathcal{L} \vec{x} &= \vec{b} \Rightarrow \vec{x} = \mathcal{L}^{-1} \vec{b} \\ \mathcal{L} \mathbf{1} &= \mathbf{I} &= \mathbf{G} \mathbf{1} \\ \mathbf{G} &= \mathcal{L}^{-1} \end{aligned}$$

10.1 1D Problems

Monday, April 20, 2026 3:16 PM

$$\mathcal{L}y = \frac{d}{dx} \left(p(x) \frac{dy}{dx} \right) + q(x)y = f(x) \quad a \leq x \leq b$$

Homogeneous BC at $a, b \Rightarrow \mathcal{L}$ is self adjoint "Hermitian"

$$\mathcal{L}G(x, t) = \delta(x - t) \Rightarrow y(x) = \int_a^b G(x, t) f(t) dt$$

Verify

$$\mathcal{L}y(x) = \mathcal{L} \int_a^b G(x, t) f(t) dt = \int_a^b \delta(x - t) f(t) dt = f(x)$$

$$\mathcal{L}y(x) = f(x)$$

Jump disc. in derivative

$$\int_{t-\varepsilon}^{t+\varepsilon} L(G(x,t)) dx = \int_{t-\varepsilon}^{t+\varepsilon} \delta(x-t) dx$$

$$\varepsilon \rightarrow 0$$

$$\int_{t-\varepsilon}^{t+\varepsilon} \left[\frac{d}{dx} \left(p(x) \frac{dG(x,t)}{dx} \right) + q(x) G(x,t) \right] dx = 1$$

$$\Rightarrow \lim_{\varepsilon \rightarrow 0} \left[\left. \frac{dG(x,t)}{dx} \right|_{x=t+\varepsilon} - \left. \frac{dG(x,t)}{dx} \right|_{x=t-\varepsilon} \right] = \frac{1}{p(t)}$$

$\frac{dG(x,t)}{dx}$ is discont. at $x=t$

Eigenfunction Expansion

$\mathcal{L}$ is Hermitian $\Rightarrow \mathcal{L}\psi = \lambda\psi$

Eigenfunctions ψ_n } $\mathcal{L}\psi_n = \lambda_n\psi_n \Rightarrow \langle \psi_m, \psi_n \rangle = \delta_{mn}$
 Eigenvalues λ_n

$$G(x,t) = \sum_{mn} g_{mn} \psi_n(x) \psi_n^*(t)$$

using $\delta(x-t) = \sum_n \psi_n(x) \psi_n^*(t)$

$$\mathcal{L}G(x,t) = \sum_{mn} g_{mn} \mathcal{L}\psi_n(x) \psi_n^*(t)$$

$$\sum_n \psi_n(x) \psi_n^*(t) = \sum_{mn} g_{mn} \lambda_n \psi_n(x) \psi_n^*(t) \Rightarrow g_{mn} = \frac{\delta_{mn}}{\lambda_n}$$

$$G(x, t) = \sum_n \frac{\varphi_n(x) \varphi_n^*(t)}{\lambda_n}$$

Symmetry of $G(x, t)$

$$G(x, t) = G^*(t, x)$$

Forms of GF

$\mathcal{L}$ with BC on (a, b)

$$\mathcal{L} G(x, t) = \delta(x - t)$$

$x < t$ $\mathcal{L} G(x, t) = 0 \Rightarrow y_1(x)$ satisfies BC at $x = a$

$$G(x, t) = y_1(x) h_1(t) \quad x < t$$

$y_2(x)$ satisfies BC at $x = b$

$x > t$

$$G(x, t) = y_2(x) h_2(t) \quad x > t$$

Symmetry $\Rightarrow$

$$G(x, t) = \begin{cases} A y_1(x) y_2(t) & , x < t \\ A y_2(x) y_1(t) & , x > t \end{cases}$$

Deriv. Jump dis.

$$A [y_1'(t) y_2(t) - y_2'(t) y_1(t)] = \frac{1}{p(t)}$$

$$y(x) = \int_a^b G(x, t) f(t) dt = \int_a^x A y_2(x) y_1(t) f(t) dt + \int_x^b A y_1(x) y_2(t) f(t) dt$$

Ex 10.1.1

Monday, April 20, 2026 3:46 PM

$$-y'' = f(x)$$

$$y(0) = y(1) = 0$$

$$p(x) = -1$$

$$y'' = 0 \Rightarrow y = c_0 + c_1 x$$

$$y_1(0) = 0 \Rightarrow y_1(x) = x$$

$$\Rightarrow y_1'(x) = 1$$

$$y_2(1) = 0 \Rightarrow y_2(x) = 1 - x$$

$$\Rightarrow y_2'(x) = -1$$

$$c_1(x, t) = \begin{cases} x(1-t), & x < t \\ (1-x)t, & x > t \end{cases}$$

$$A[-t - (1-t)] = \frac{1}{-1} \Rightarrow A = 1$$

$$\underline{f(x) = \sin \pi x} \Rightarrow y(x) = \int_0^x (1-x)t \sin \pi t \, dt + \int_x^1 x(1-t) \sin \pi t \, dt = \frac{1}{\pi^2} \sin \pi x$$

$$\underline{f(x) = \cos \pi x} \Rightarrow y(x) = \int_0^x (1-x)t \cos \pi t \, dt + \int_x^1 x(1-t) \cos \pi t \, dt = \frac{1}{\pi^2} (2x-1 + \cos \pi x)$$

Ex 10.1.2

$$\mathcal{L}y = y'' + y = f(x)$$

$$y(0) = 0$$

$$y'(0) = 0$$

$$p(x) = 1$$

$$\mathcal{L}y = 0 \Rightarrow y_1 = \sin x, \quad y_2 = \cos x$$

$$\underline{x < t}$$

IV conditions $\Rightarrow$

$$G(x, t) = 0$$

$$\underline{x > t}$$

$$G(x, t) = c_1(t) \sin x + c_2(t) \cos x$$

$$G(t_-, t) = G(t_+, t) \Rightarrow 0 = c_1(t) \sin t + c_2(t) \cos t$$

$$\frac{dG}{dx}(t_+, t) - \frac{dG}{dx}(t_-, t) = \frac{1}{p(t)} \Rightarrow 1 = c_1(t) \cos t - c_2(t) \sin t = 0$$

$$G(x, t) = \cos t \sin x - \sin t \cos x = \sin(x - t)$$

$$G(x, t) = \begin{cases} 0 & , x < t \\ \sin(x - t) & , x > t \end{cases}$$

$$y(x) = \int_0^{\infty} G(x, t) f(t) dt = \int_0^x \sin(x - t) f(t) dt$$

Ex 10.1.3 BC at ∞

Monday, April 20, 2026 4:08 PM

$$\left(\frac{d^2}{dx^2} + k^2\right) \psi(x) = f(x)$$

$$p(x) = 1$$

Wave traveling
to $+\infty$

$$e^{i(kx - \omega t)}$$

Outgoing wave ($e^{i\omega t}$)

Homogeneous $\eta \Rightarrow y_1 = e^{ikx}$

$$y_2 = e^{-ikx}$$

$$G(x, x') = \begin{cases} A y_1(x') y_2(x) & x > x' \\ A y_2(x') y_1(x) & x < x' \end{cases}$$

$$x > x'$$

$$x < x'$$

$$A = \frac{1}{y_1'(x) y_2(x) - y_1(x) y_2'(x)} = \frac{1}{ik + ik} = \frac{-i}{2k}$$

$$G(x, x') = -\frac{i}{2k} \exp(i|x - x'|)$$

10.2 Problems in 2D & 3D

Wednesday, April 22, 2026 2:50 PM

Basic features

1) Homog. PDE $\mathcal{L}\psi(\vec{r}_1) = 0$ & BCs define $G(\vec{r}_1, \vec{r}_2)$

$$\mathcal{L}G(\vec{r}_1, \vec{r}_2) = \delta(\vec{r}_1 - \vec{r}_2)$$

2) Inhomog. PDE $\mathcal{L}\psi(\vec{r}_1) = f(\vec{r}_1)$

$$\Rightarrow \psi(\vec{r}_1) = \int G(\vec{r}_1, \vec{r}_2) f(\vec{r}_2) d^3r_2$$

3) Hermitian eigenvalue problem $\mathcal{L}\psi = \lambda\psi \Rightarrow \phi_n(\vec{r}_1)$ & λ_n

$$* G(\vec{r}_1, \vec{r}_2) = \sum_n \frac{\phi_n(\vec{r}_1) \phi_n^*(\vec{r}_2)}{\lambda_n}$$

$$* \text{Symmetry} \quad G(\vec{r}_1, \vec{r}_2) = G^*(\vec{r}_2, \vec{r}_1)$$

4) $G(\vec{r}_1, \vec{r}_2)$ continuous except at $\vec{r}_1 = \vec{r}_2$

Self adjoint Problem

Wednesday, April 22, 2026 3:08 PM

$$\mathcal{L}\psi(\vec{r}) = \vec{\nabla} \cdot [\mu(\vec{r}) \vec{\nabla} \psi(\vec{r})] + q(\vec{r}) \psi(\vec{r}) = f(\vec{r})$$

$$\langle q(\vec{r}, \vec{r}_1) | \mathcal{L} G(\vec{r}, \vec{r}_2) \rangle = \langle \mathcal{L} G(\vec{r}, \vec{r}_1) | G(\vec{r}, \vec{r}_2) \rangle$$

$$\langle G(\vec{r}, \vec{r}_1) | \delta(\vec{r} - \vec{r}_2) \rangle = \langle \delta(\vec{r} - \vec{r}_1) | G(\vec{r}, \vec{r}_2) \rangle$$

$$G^*(\vec{r}_2, \vec{r}_1) = G(\vec{r}_1, \vec{r}_2)$$

Eigenfunction expansion

$$\mathcal{L} G(\vec{r}_1, \vec{r}_2) = \delta(\vec{r}_1 - \vec{r}_2) \Rightarrow G(\vec{r}_1, \vec{r}_2) = \sum_n \frac{\phi_n(\vec{r}_1) \phi_n^*(\vec{r}_2)}{\lambda_n}$$

$$\mathcal{L} G(\vec{r}_1, \vec{r}_2) - \tau G(\vec{r}_1, \vec{r}_2) = \delta(\vec{r}_1 - \vec{r}_2) \Rightarrow G(\vec{r}_1, \vec{r}_2) = \sum_n \frac{\phi_n(\vec{r}_1) \phi_n^*(\vec{r}_2)}{\lambda_n - \tau}$$

Form of Green Function

Wednesday, April 22, 2026 3:19 PM

Laplace Eqn 3D

$$\nabla^2 G(\vec{r}_1, \vec{r}_2) = \delta(\vec{r}_1 - \vec{r}_2)$$

$$\lim_{\vec{r}_1 \rightarrow \infty} G(\vec{r}_1, \vec{r}_2) = 0$$

Sph. Symmetry $G \sim |\vec{r}_1 - \vec{r}_2| = r_{12}$

Step 1 Integration

$$\int_{r_{12} < a} \vec{\nabla} \cdot \vec{\nabla} G(\vec{r}_1, \vec{r}_2) d^3 r_1 = 1$$

$$G(\vec{r}_1, \vec{r}_2) = -\frac{1}{4\pi} \frac{1}{|\vec{r}_1 - \vec{r}_2|}$$

Step 2 Div. Thm

$$\int_{r_{12}=a} \vec{\nabla} G(\vec{r}_1, \vec{r}_2) \cdot d\vec{\sigma}_1 = 4\pi a^2 \left. \frac{dG}{dr_{12}} \right|_{r_{12}=a} = 1 \Rightarrow \frac{dG}{dr_{12}} = \frac{1}{4\pi r_{12}^2}$$

Laplace Eqn 2D

Wednesday, April 22, 2026 3:28 PM

$$\int_{\rho_{12}=a} \vec{\nabla}_1 G(\vec{\rho}_1, \vec{\rho}_2) \cdot d\vec{\sigma}_1 = 2\pi a \left. \frac{dG}{d\rho_{12}} \right|_{\rho_{12}=a} = 1$$

$$\frac{dG}{d\rho_{12}} = \frac{1}{2\pi \rho_{12}} \Rightarrow G(\vec{\rho}_1, \vec{\rho}_2) = \frac{1}{2\pi} \ln |\vec{\rho}_1 - \vec{\rho}_2|$$

Table 10.1 Fundamental Green's Functions^a

	Laplace ∇^2	Helmholtz ^b $\nabla^2 + k^2$	Modified Helmholtz ^c $\nabla^2 - k^2$
1-D	$\frac{1}{2} x_1 - x_2 $	$-\frac{i}{2k} \exp(ik x_1 - x_2)$	$-\frac{1}{2k} \exp(-k x_1 - x_2)$
2-D	$\frac{1}{2\pi} \ln \rho_1 - \rho_2 $	$-\frac{i}{4} H_0^{(1)}(k \rho_1 - \rho_2)$	$-\frac{1}{2\pi} K_0(k \rho_1 - \rho_2)$
3-D	$-\frac{1}{4\pi} \frac{1}{ \mathbf{r}_1 - \mathbf{r}_2 }$	$-\frac{\exp(ik \mathbf{r}_1 - \mathbf{r}_2)}{4\pi \mathbf{r}_1 - \mathbf{r}_2 }$	$-\frac{\exp(-k \mathbf{r}_1 - \mathbf{r}_2)}{4\pi \mathbf{r}_1 - \mathbf{r}_2 }$

^a Boundary conditions: For the Helmholtz equation, outgoing wave;
for modified Helmholtz and 3-D Laplace equations, $G \rightarrow 0$ at infinity;
for 1-D and 2-D Laplace equation, arbitrary.

^b $H_0^{(1)}$ is a Hankel function, Section 14.4.

^c K_0 is a modified Bessel function, Section 14.5.

$$\frac{1}{|\vec{r}_1 - \vec{r}_2|} = \sum_{l=0}^{\infty} \frac{4\pi}{2l+1} \frac{r_l^l}{r_s^{l+1}} \sum_{m=-l}^{m=l} Y_l^m(\mu_1) Y_l^{m*}(\mu_2)$$

Ex 10.2.2

Wednesday, April 22, 2026 3:38 PM

Born approximation - QM scattering

$$E = \frac{\hbar^2 k^2}{2m}$$

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\vec{r}) + V(\vec{r}) \psi(\vec{r}) = E \psi(\vec{r})$$

$$\nabla^2 \psi(\vec{r}) + k^2 \psi(\vec{r}) = \left[\frac{2m}{\hbar^2} V(\vec{r}) \psi(\vec{r}) \right]$$

Asymptotic form: $\psi(\vec{r}) \sim \underbrace{e^{i\vec{k}_0 \cdot \vec{r}}}_{\text{incident wave}} + f_k(\theta, \varphi) \underbrace{\frac{e^{ikr}}{r}}_{\text{spherical wave}}$

$$|f_k(\theta, \varphi)|^2 \sim \text{cross section}$$

Green's function

$$\psi(\vec{r}_1) = \int \frac{d^3 r_2}{h^2} V(\vec{r}_2) \underbrace{G(\vec{r}_1, \vec{r}_2)}_{-\frac{e^{i\vec{k} \cdot \vec{r}}}{r}} \psi(\vec{r}_2) d^3 r_2$$

Exact

Lippmann-Schwinger
eqn

$$\psi(\vec{r}_1) = e^{i\vec{k}_0 \cdot \vec{r}} - \int \frac{d^3 r_2}{h^2} V(\vec{r}_2) \frac{e^{i\vec{k} \cdot (\vec{r}_1 - \vec{r}_2)}}{4\pi|\vec{r}_1 - \vec{r}_2|} \psi(\vec{r}_2) d^3 r_2$$

Born approx. $\psi(\vec{r}_2) \sim e^{i\vec{k}_0 \cdot \vec{r}_2}$

$$\psi(\vec{r}_1) = e^{i\vec{k}_0 \cdot \vec{r}} - \int \frac{d^3 r_2}{h^2} V(\vec{r}_2) \frac{e^{i\vec{k} \cdot (\vec{r}_1 - \vec{r}_2)}}{4\pi|\vec{r}_1 - \vec{r}_2|} e^{i\vec{k}_0 \cdot \vec{r}_2} d^3 r_2$$

$$\sim e^{i\vec{k}_0 \cdot \vec{r}} + f_k(\theta, \phi) \frac{e^{i\vec{k} \cdot \vec{r}}}{r}$$

General Born approx.

Wednesday, April 22, 2026 3:53 PM

$$\mathcal{L}|\psi\rangle = 0 \Rightarrow |\psi\rangle = |\phi\rangle$$

$$\mathcal{L}G_0(\vec{r}, \vec{r}') = \delta(\vec{r}, \vec{r}')$$

$$\mathcal{L}|\psi\rangle = V|\psi\rangle$$

$$|\psi\rangle = |\phi\rangle + G_0 V |\psi\rangle \quad \text{L-S eqn}$$

$$|\psi\rangle = |\phi\rangle + G_0 V |\psi\rangle$$

$$|\psi\rangle = |\phi\rangle + G_0 V |\phi\rangle + (G_0 V)^2 |\phi\rangle + (G_0 V)^3 |\phi\rangle + \dots$$

Born series

$$= [1 - G_0 V]^{-1} |\phi\rangle$$